S&S QUIZ- 8 SOLUTION

Sol(1):- Given causal LTI system's difference eqn,

$$y[n] + \frac{1}{2}y[n-1] = x[n] \quad \text{—— (1)}$$

(a) As we know, $x[n] \underset{IDFT}{\overset{DFT}{\rightleftharpoons}} X(e^{j\omega})$

where, $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] \cdot e^{-j\omega n}$

$$x[n] = \frac{1}{2\pi}\int_{2\pi} X(e^{j\omega}) \cdot e^{j\omega n} \cdot d\omega \quad \text{—— (2)}$$

Replacing $n \rightarrow (n-1)$ in eqn(2)'s both side, we get –

$$x[n-1] = \frac{1}{2\pi}\int_{2\pi} X(e^{j\omega}) \cdot e^{j\omega(n-1)} \cdot d\omega$$

$$x[n-1] = \frac{1}{2\pi}\int_{2\pi} e^{-j\omega} \cdot X(e^{j\omega})\, e^{j\omega n}\, d\omega$$

$$\Rightarrow \quad x[n-1] \underset{IDFT}{\overset{DFT}{\rightleftharpoons}} e^{-j\omega} \cdot X(e^{j\omega}) \quad \text{—— (3)}$$ (1 Mark)

Taking DFT of both side of eqn(1) with the help of eqn(3),

$$Y(e^{j\omega}) + \frac{1}{2}e^{-j\omega} \cdot Y(e^{j\omega}) = X(e^{j\omega})$$

$$Y(e^{j\omega})\left[1 + \frac{1}{2}e^{-j\omega}\right] = X(e^{j\omega})$$

$$\frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{1}{\left(1+\frac{1}{2}e^{-j\omega}\right)}$$

$$\therefore H(e^{j\omega}) = \frac{1}{\left(1+\frac{1}{2}e^{-j\omega}\right)} \rightarrow \text{(2 Marks)}$$

(b) Given input, $x[n] = \left(-\frac{1}{2}\right)^n u[n]$ —— (4)

As we know, $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] \cdot e^{-j\omega n} \rightleftharpoons x[n]$

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$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} \left(-\frac{1}{2}\right)^n u[n]\, e^{-j\omega n}$$

$$= \sum_{n=0}^{\infty} \left(-\frac{1}{2} e^{-j\omega}\right)^n$$

$$= \frac{1}{\left(1 + \frac{1}{2} e^{-j\omega}\right)} \quad \text{— (5)}$$

→ (2 Marks)

(c) Given input DFT, $X(e^{j\omega}) = 1 + 2e^{-3j\omega}$ — (7)

By part (a), we know —

$$\frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{1}{\left(1 + \frac{1}{2} e^{-j\omega}\right)}$$

$$\therefore Y(e^{j\omega}) = \frac{\left(1 + 2e^{-j3\omega}\right)}{\left(1 + \frac{1}{2} e^{-j\omega}\right)}$$

$$= \frac{1}{\left(1 + \frac{1}{2} e^{-j\omega}\right)} + \frac{2e^{-j3\omega}}{\left(1 + \frac{1}{2} e^{-j\omega}\right)} \quad \text{— (8)}$$

→ (1 Marks)

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Taking IDFT of eqn (8) with the help of eqn (3) & eqn (5),

$$y[n] = \left(-\frac{1}{2}\right)^n u[n] + 2\left(-\frac{1}{2}\right)^{n-3} u[n-3]$$

→ (2 Marks)

Part (b) remaining part:

By part (a), we know –

$$\frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{1}{\left(1+\frac{1}{2}e^{-j\omega}\right)}$$

$$Y(e^{j\omega}) = \frac{1}{\left(1+\frac{1}{2}e^{-j\omega}\right)^2} \quad \text{—— (6)}$$

By eqn (5), we get –

As we know that –

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]\cdot e^{-j\omega n}$$

$$\frac{d\,X(e^{j\omega})}{d\omega} = (-j)\sum_{n=-\infty}^{\infty} n x[n]\cdot e^{-j\omega n}$$

$$j\frac{d\,X(e^{j\omega})}{d\omega} = \sum_{n=-\infty}^{\infty} n x[n]\cdot e^{-j\omega n}$$

$$a^n u[n] \underset{IDFT}{\overset{DFT}{\rightleftharpoons}} \frac{1}{(1-ae^{-j\omega})}$$

$$n\,a^n u[n] \underset{IDFT}{\overset{DFT}{\rightleftharpoons}} j\frac{d}{d\omega}\left[\frac{1}{1-ae^{-j\omega}}\right]$$

$$(n+1)\,a^n u[n+1] \underset{IDFT}{\overset{DFT}{\rightleftharpoons}} \frac{1}{(1-ae^{-j\omega})^2} \quad \text{— (7)}$$

→ (1 Marks)

Taking IDFT of eqn (6) by using eqn (7), we get –

$$y[n] = (n+1)\left(-\frac{1}{2}\right)^n u[n+1]$$

→ (1 Marks)

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Taking IDFT of eqn (8) with the help of eqn (3) & eqn (5),

$$y[n] = \left(-\frac{1}{2}\right)^n u[n] + 2\left(-\frac{1}{2}\right)^{n-3} u[n-3]$$

→ (2 Marks)

Part (b) remaining part:

By part (a), we know –

$$\frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{1}{\left(1+\frac{1}{2}e^{-j\omega}\right)}$$

$$Y(e^{j\omega}) = \frac{1}{\left(1+\frac{1}{2}e^{-j\omega}\right)^2} \quad \text{—— (6)}$$

By eqn (5), we get –

As we know that –

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] \cdot e^{-j\omega n}$$

$$\frac{d}{d\omega} X(e^{j\omega}) = (-j) \sum_{n=-\infty}^{\infty} n x[n] \cdot e^{-j\omega n}$$

$$j \frac{d}{d\omega} X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} n x[n] \cdot e^{-j\omega n}$$

$$a^n u[n] \underset{IDFT}{\overset{DFT}{\rightleftharpoons}} \frac{1}{(1-ae^{-j\omega})}$$

$$n a^n u[n] \underset{IDFT}{\overset{DFT}{\rightleftharpoons}} j \frac{d}{d\omega}\left[\frac{1}{1-ae^{-j\omega}}\right]$$

$$(n+1) a^n u[n] \underset{IDFT}{\overset{DFT}{\rightleftharpoons}} \frac{1}{(1-ae^{-j\omega})^2} \quad \text{—— (7)}$$

→ (1 Marks)

Taking IDFT of eqn (6) by using eqn (7), we get –

$$y[n] = (n+1)\left(-\frac{1}{2}\right)^n u[n]$$

→ (1 Marks)

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